

# Vishal Gaur

Tampere, Finland | +358 41 716 3235 | [LinkedIn](#) | [Portfolio](#) | [Google Scholar](#)

## Summary

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Computer Scientist with 5+ years of hands-on experience building machine learning solutions using generative AI in a collaborative environment. Currently developing transformer-based foundation models at NASA, with expertise in end-to-end AI solution ownership — from data pipeline construction and model training on HPC clusters to assisting automated cloud deployment and monitoring. Proven ability to collaborate across technical and business teams to turn complex requirements into reliable AI systems. Experienced in building automation workflows, API-driven services, and reusable data engineering components.

## Technical Skills

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**Languages:** Python, SQL, C#, C++

**AI/ML Frameworks:** PyTorch, Hugging Face, Transformers, scikit-learn, SciPy, chromaDB, pgvector, OpenCV, YOLO

**Generative AI & Architectures:** Transformer models (encoder/decoder), foundation model fine-tuning, LoRA (parameter-efficient adaptation), LLM, RAGs, distributed model training (DDP), Model quantization

**Data & Optimization:** ML Optimization, MLOps, STAC metadata pipelines

**Cloud & Infrastructure:** AWS (S3, EC2, Lambda, Athena, Sagemaker), WandB, Docker, SLURM, PBS

**APIs & Integration:** FastAPI, RESTful services, API Gateway, cloud-native service integration

**Platforms & Tools:** Linux, macOS, QGIS, GitHub, Claude code, GPT codex, Prompt Engineering

**Domains:** Foundation Models, Computer Vision, LLMs, RAGs, Intelligent Automation, Recommendation Systems, Geospatial Analytics, Explainable AI

## Professional Experience

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### Computer Scientist

Feb 2023 – Present

*NASA IMPACT, Huntsville, AL, USA*

#### Weather & Climate Foundation Model (Prithvi WxC)

- Designed and executed end-to-end data processing pipelines for petabyte-scale reanalysis time series datasets, enabling reliable model training across isolated computing environments.
- Applied LoRA-based parameter-efficient fine-tuning on pre-trained transformer backbones to create task-specific downstream models, reducing training cost while maintaining prediction quality.

#### Helio Foundation Model (Surya)

- Led exploratory data analysis on multi-year solar image archives, identifying data quality issues and informing architecture choices for the generative model pipeline.
- Trained and benchmarked multiple transformer architectures on distributed compute systems like NVIDIA DGX and NAS clusters, managing distributed workflows via SLURM and PBS schedulers.
- Fine-tuned Foundation model using lora adapters and tasks specific decoders, to predict solar flares, solar wind and assisted the validation of model results with the science team.
- Designed various convolution based architectures to create baselines for solar prediction and solar wind prediction.

#### Harmonized Landsat and Sentinel (HLS) Bandpass Research

- Built a clean, reproducible dataset by aggregating imagery from multiple remote sensing sources, establishing standardized data preparation patterns reusable across teams.
- Created deep learning model benchmarks and systematically evaluated performance across architectures, documenting best practices and methodology comparisons for broader team adoption.

#### US Greenhouse Gas Center

- Automated end-to-end transformation pipelines converting geospatial data to Cloud-Optimized GeoTIFFs (COGs), eliminating manual processing steps and improving data delivery speed.
- Orchestrated data ingestion automation workflows using Apache Airflow, cataloging outputs into STAC-compliant metadata stores for scalable discovery and access.
- Managed and monitored AWS cloud infrastructure (Lambda, API Gateway, EC2) supporting the platform backend. Contributed to a nationally recognized project presented at the White House and COP28.

### Data Scientist

Apr 2022 – Jan 2023

*Sanoma Media Finland Oy, Helsinki, Finland*

- Developed and deployed a production recommendation engine using ALS collaborative filtering within the Sanoma Unified Recommendation Framework, directly integrated with content delivery systems.

- Collaborated with business stakeholders to identify automation and AI opportunities, translating real business needs into multiple machine learning proof-of-concept prototypes.
- Refactored legacy Scala data pipelines into Python, creating reusable components that improved maintainability and accelerated iteration for the data science team.
- Performed large-scale exploratory data analysis on user engagement data collected across multiple digital media platforms, surfacing actionable insights for editorial and product teams.

#### Data Science Trainee

Jun 2021 – Sep 2021

*Kone Corporation, Espoo, Finland*

- Analyzed IoT sensor telemetry from elevator and escalator systems to identify anomalies and patterns, delivering insights that led to measurable improvements in device monitoring services.
- Built and executed cloud-based data analysis workflows on AWS (Athena, S3, EC2) to process large-scale industrial sensor datasets efficiently.

#### Data Science Trainee – Thesis Worker

Jun 2019 – May 2021

*Nokia Technologies, Espoo, Finland*

- Owned the full AI solution lifecycle: constructed a novel annotated dataset for patent office actions, developed a text classification model, designed a cloud architecture for deployment, and built a .NET desktop prototype for end-user access.
- Delivered a production-ready cloud solution integrating the ML model as a service, demonstrating end-to-end ownership from data collection through deployment and monitoring.

### Education

**M.Sc. in Information Technology** (Data Engineering and Machine Learning)

2018 – 2020

*Tampere University, Tampere, Finland*

Full tuition fee waiver scholarship (13,000 EUR/year) awarded for academic merit.

**B.Tech in Computer Science Engineering**

2013 – 2017

*Manav Rachna International University, Faridabad, India*

### Selected Publications and Research

- **Prithvi WxC: Foundation Model for Weather and Climate** – Developed and evaluated a transformer-based foundation model in collaboration with NASA, IBM, Oak Ridge National Laboratory, and NVIDIA.
- **Surya: Foundation Model for Heliophysics** – Co-authored research on training large-scale generative architectures for solar physics applications.
- **ML Global Simulation of Nonlocal Gravity Wave Propagation** – Contributed ML-based simulation methods for atmospheric gravity wave modeling.
- Led a hands-on workshop at INGARSS and Colorado State University on fine-tuning foundation models using AWS Sagemaker, guiding 50+ participants through practical implementation patterns and best practices.
- Cross-organizational collaboration with NASA, IBM, Oak Ridge Labs, NVIDIA, Arizona State University, Georgia State University, and Colorado State University.

### Grants and Awards

- **MSME Enterprise Grant** – Awarded 7,000 EUR for the Sparsh entrepreneurial initiative.
- **Dentist Posture Belt Innovation Grant** – Received 1,650 EUR for a health-tech hardware prototype.
- **Full Tuition Scholarship** – Tampere University, two academic years (26,000 EUR total).